

Bitwise linear trail through **LBlock** found by SAT

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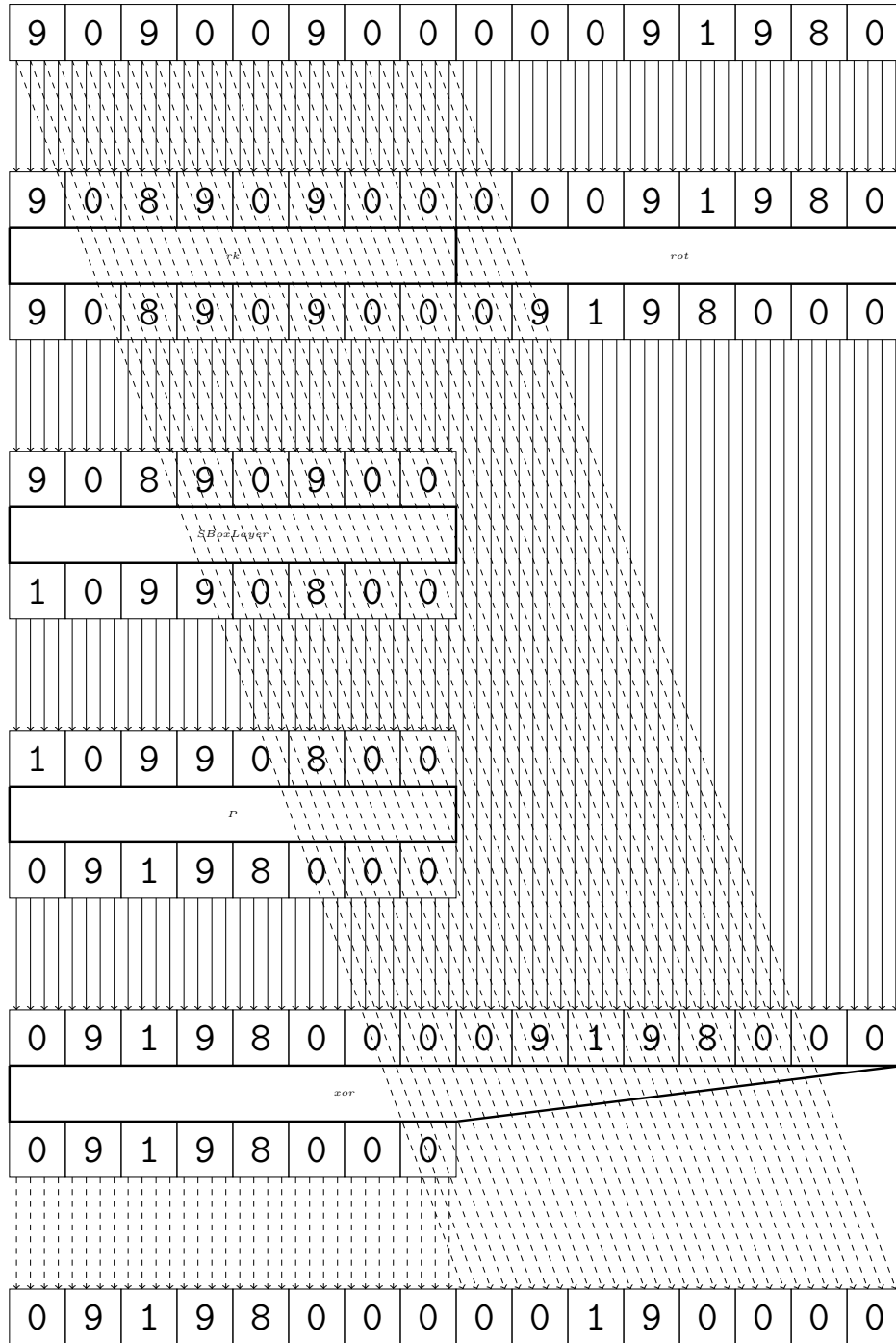
1 LBlock

Maximal linear correlation: 2^{-22}

9	0	9	0	0	9	0	0	0	0	9	1	9	8	0
9	0	9	0	0	9	0	0	0	0	9	1	9	8	0
LBlock round 1														
0	9	1	9	8	0	0	0	0	0	1	9	0	0	0
0	9	1	9	8	0	0	0	0	0	1	9	0	0	0
LBlock round 2														
1	9	0	0	0	0	0	0	0	0	1	0	8	0	0
1	9	0	0	0	0	0	0	0	0	1	0	8	0	0
LBlock round 3														
1	0	8	0	0	0	0	0	0	0	0	0	0	0	0
1	0	8	0	0	0	0	0	0	0	0	0	0	0	0
LBlock round 4														
0	0	0	0	0	0	0	0	0	1	0	8	0	0	0
0	0	0	0	0	0	0	0	0	1	0	8	0	0	0
LBlock round 5														
8	0	0	0	0	0	1	0	0	8	0	0	b	0	0
8	0	0	0	0	0	1	0	0	8	0	0	b	0	0
LBlock round 6														
0	0	b	0	0	0	0	8	0	0	0	0	0	0	0
0	0	b	0	0	0	0	8	0	0	0	0	0	0	0
LBlock round 7														
0	0	0	0	0	0	0	0	0	0	b	0	0	0	8
0	0	0	0	0	0	0	0	0	0	b	0	0	0	8
LBlock round 8														
b	0	0	0	0	8	0	0	0	5	0	0	0	0	1
b	0	0	0	0	8	0	0	0	5	0	0	0	0	1
LBlock round 9														
0	0	0	0	0	1	0	5	b	0	0	0	0	8	4
0	0	0	0	0	1	0	5	b	0	0	0	0	8	4
LBlock round 10														
0	0	0	8	4	b	b	0	0	0	1	0	c	0	0
0	0	0	8	4	b	b	0	0	0	1	0	c	0	0
LBlock round 11														
1	0	c	0	0	0	0	0	4	a	0	8	4	b	b
1	0	c	0	0	0	0	0	4	a	0	8	4	b	b
Final step														
4	a	0	8	4	b	b	0	1	0	c	0	0	0	0
4	a	0	8	4	b	b	0	1	0	c	0	0	0	0

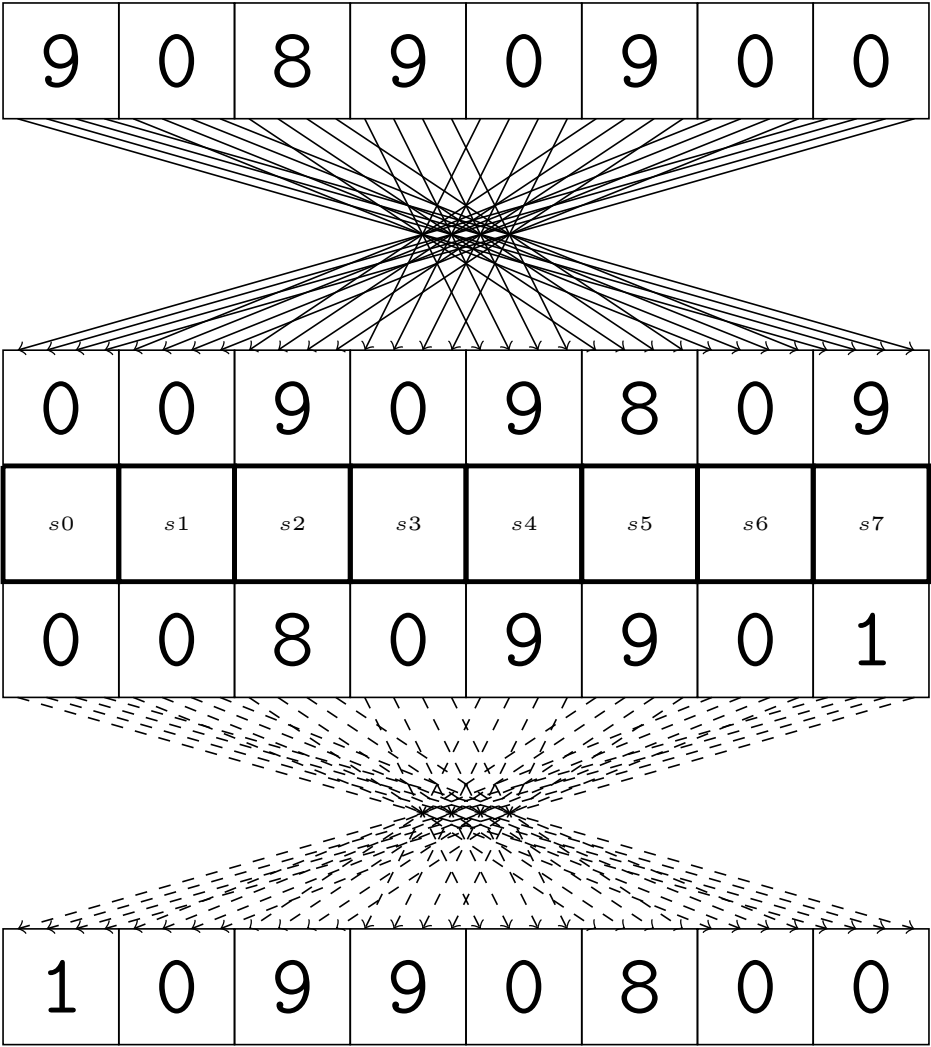
2 LBlock_round

Maximal linear correlation: $2^{-4.0}$

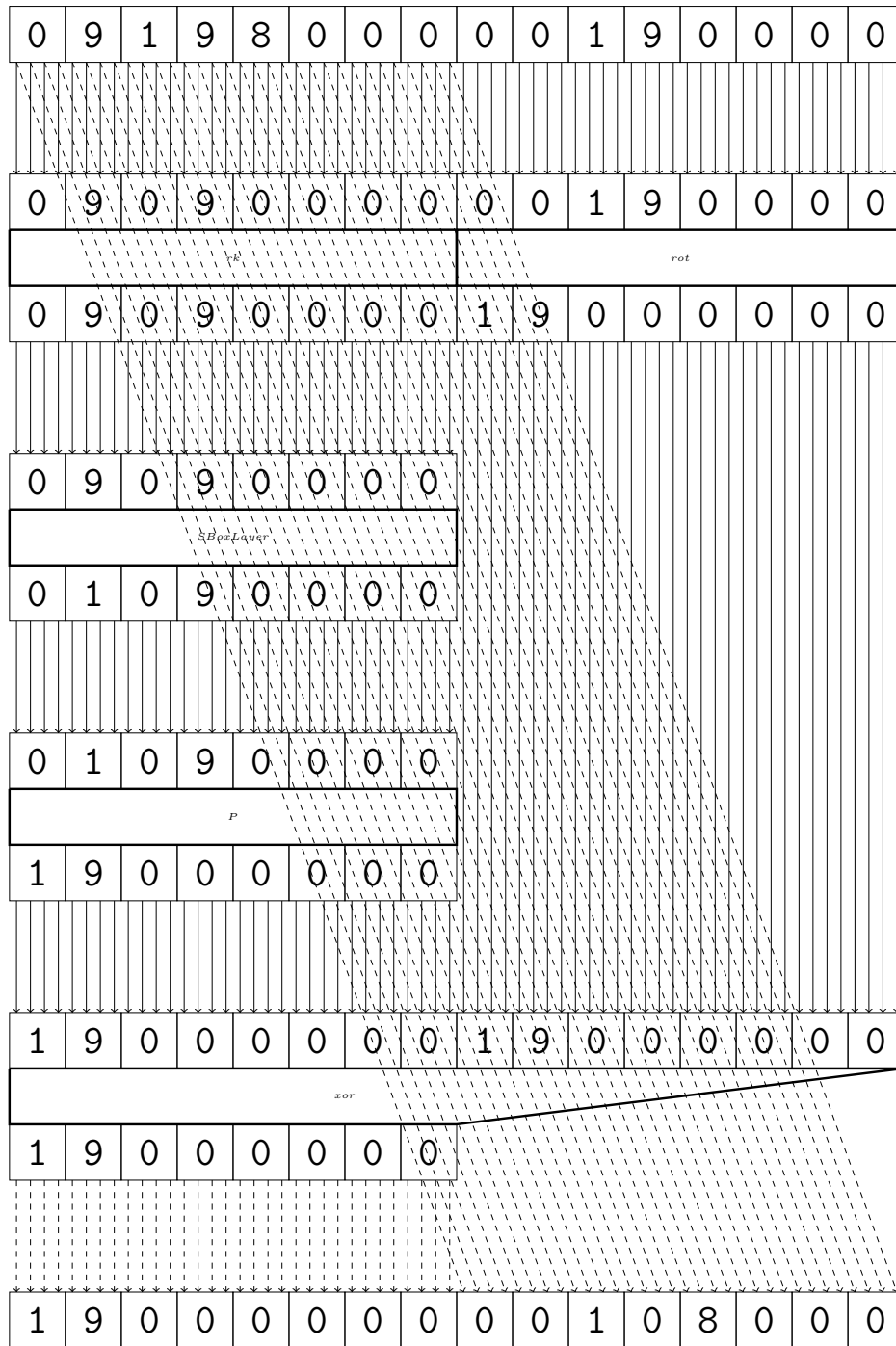


3 SBoxLayer

Maximal linear correlation: $2^{-4.0}$

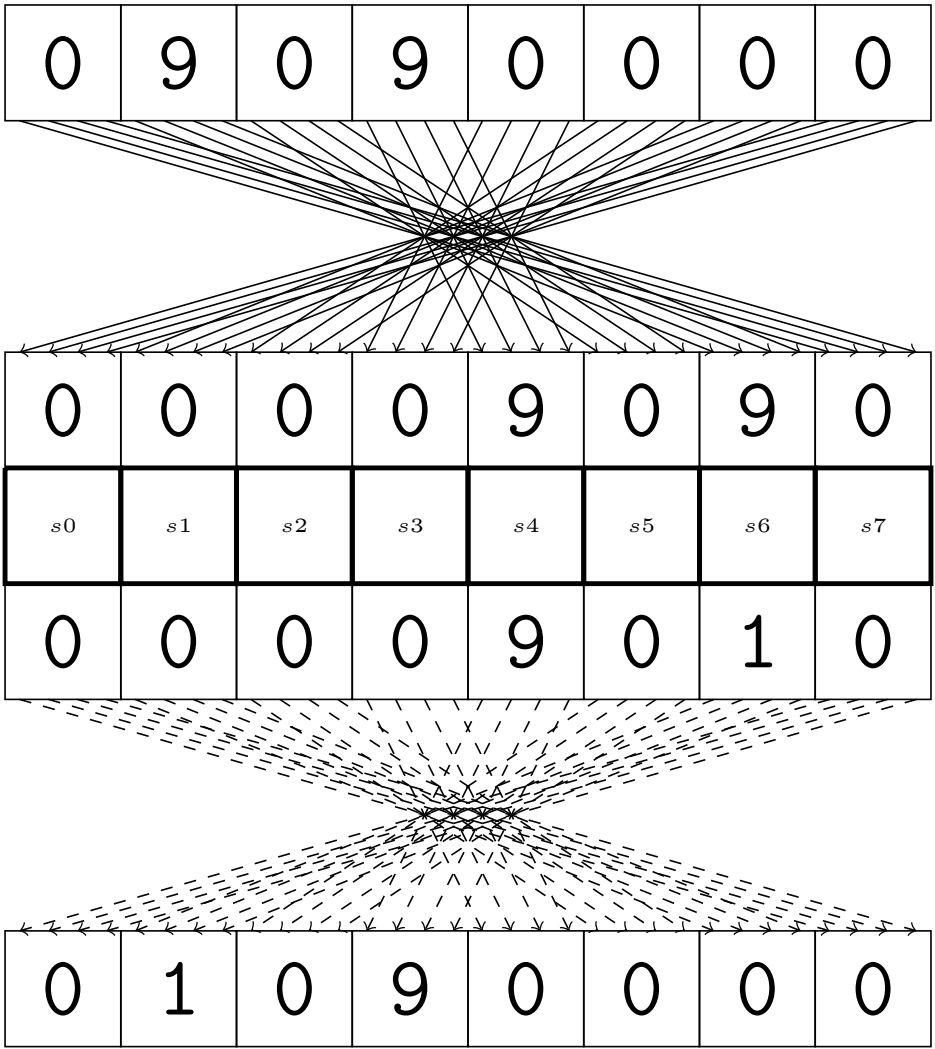


4 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

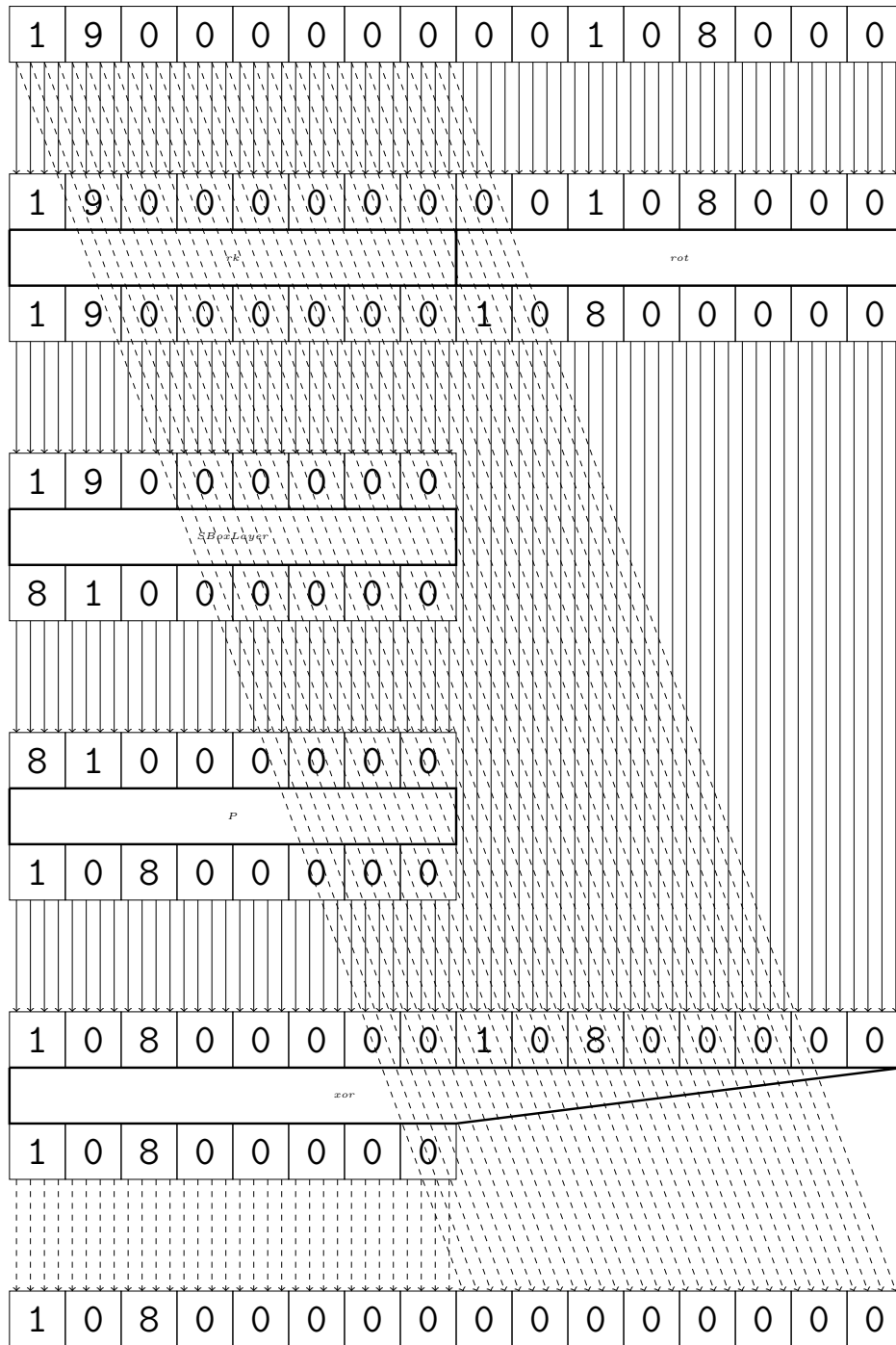
5 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



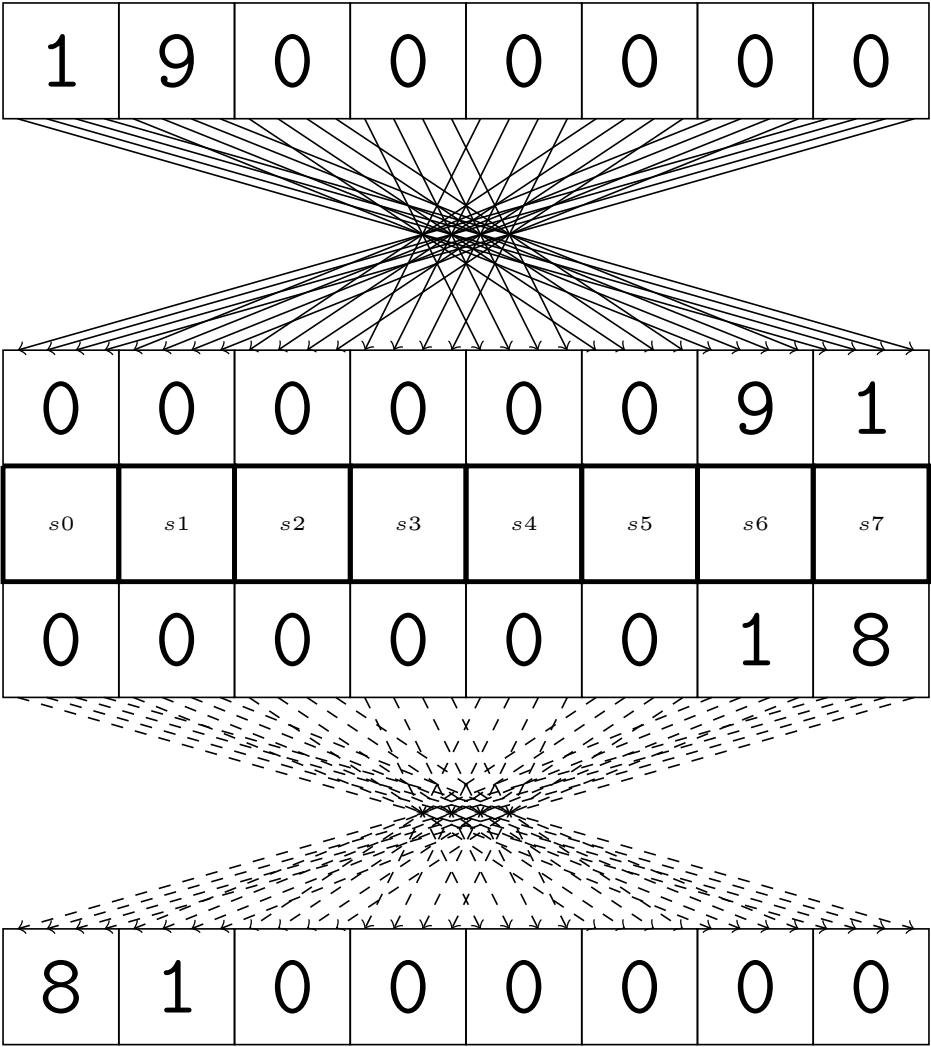
6 LBlock_round

Maximal linear correlation: $2^{-2.0}$

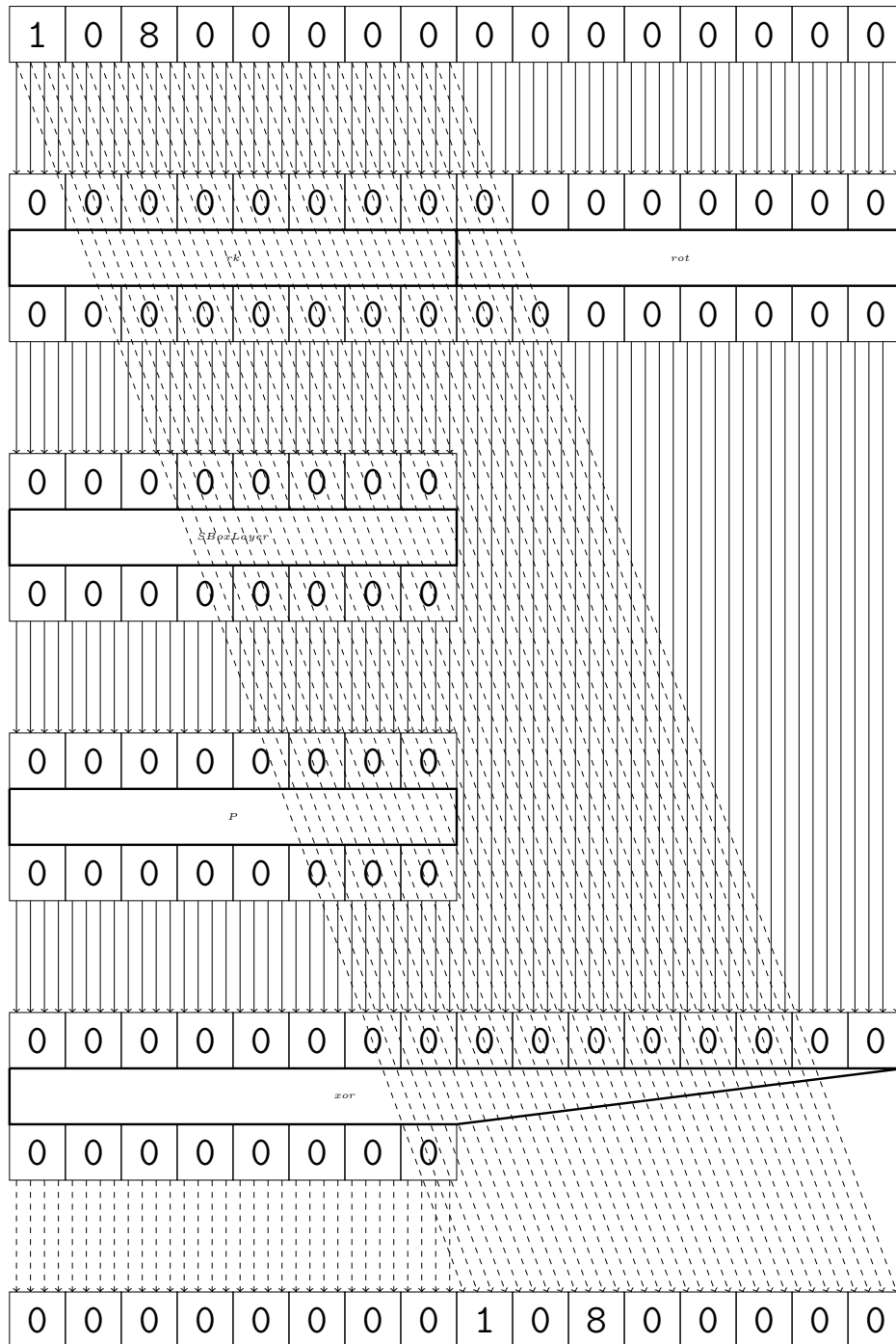


7 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

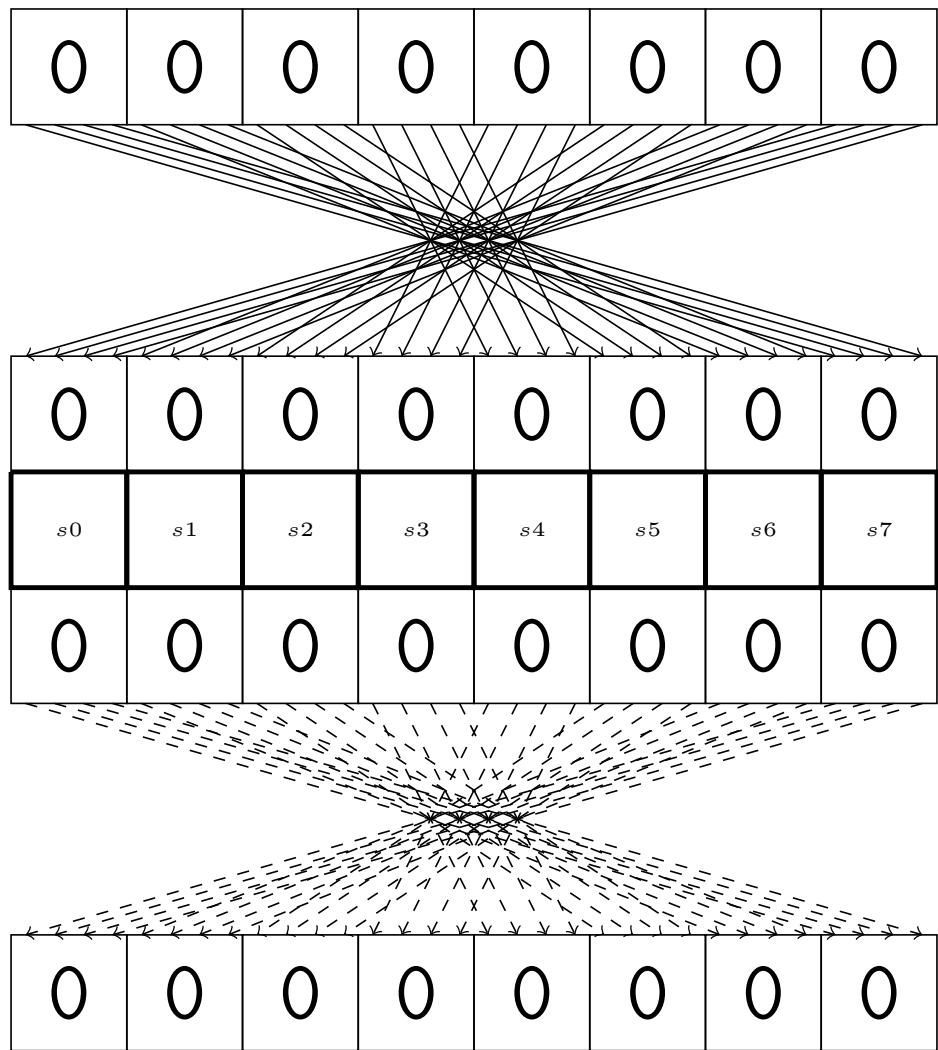


8 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

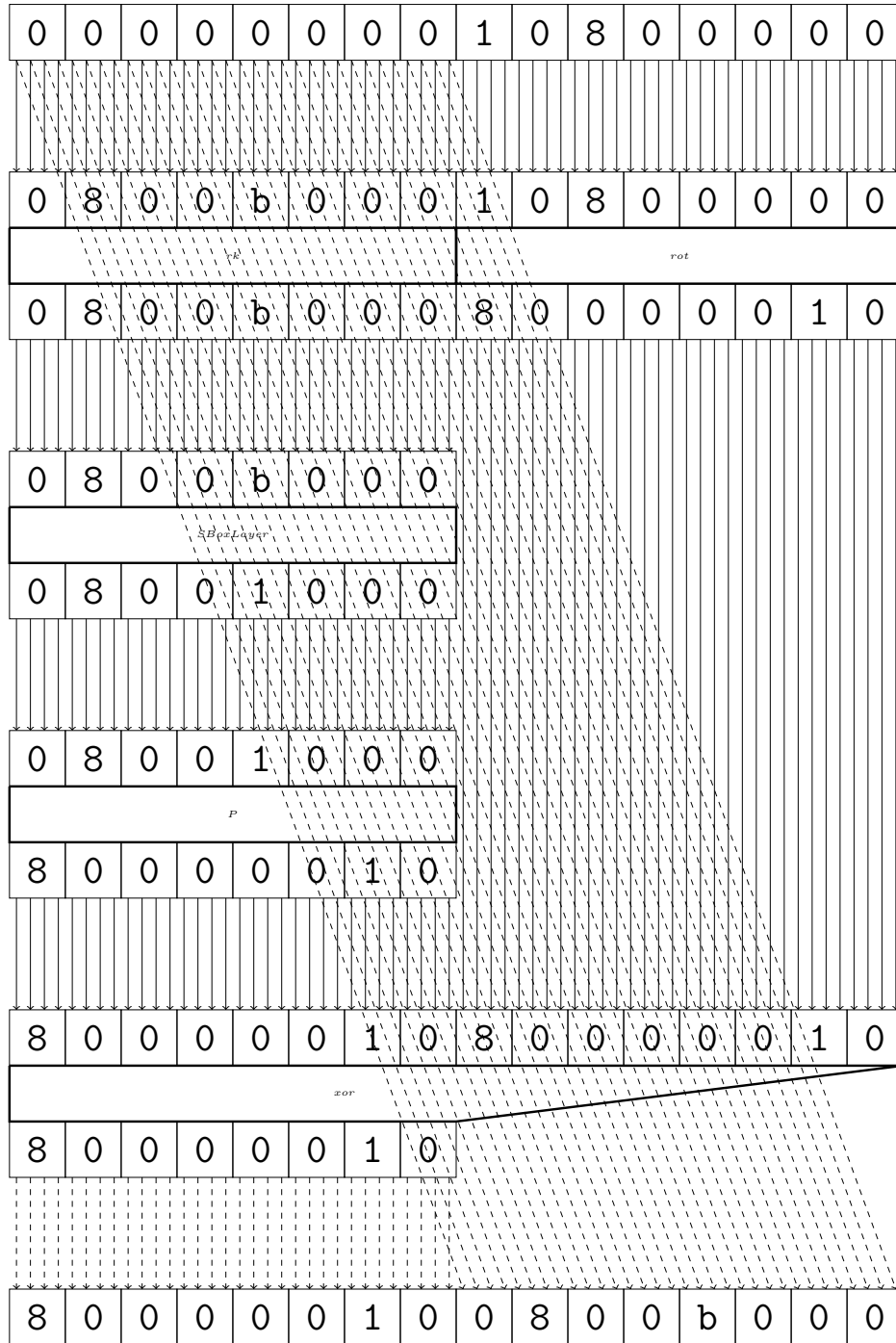
9 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



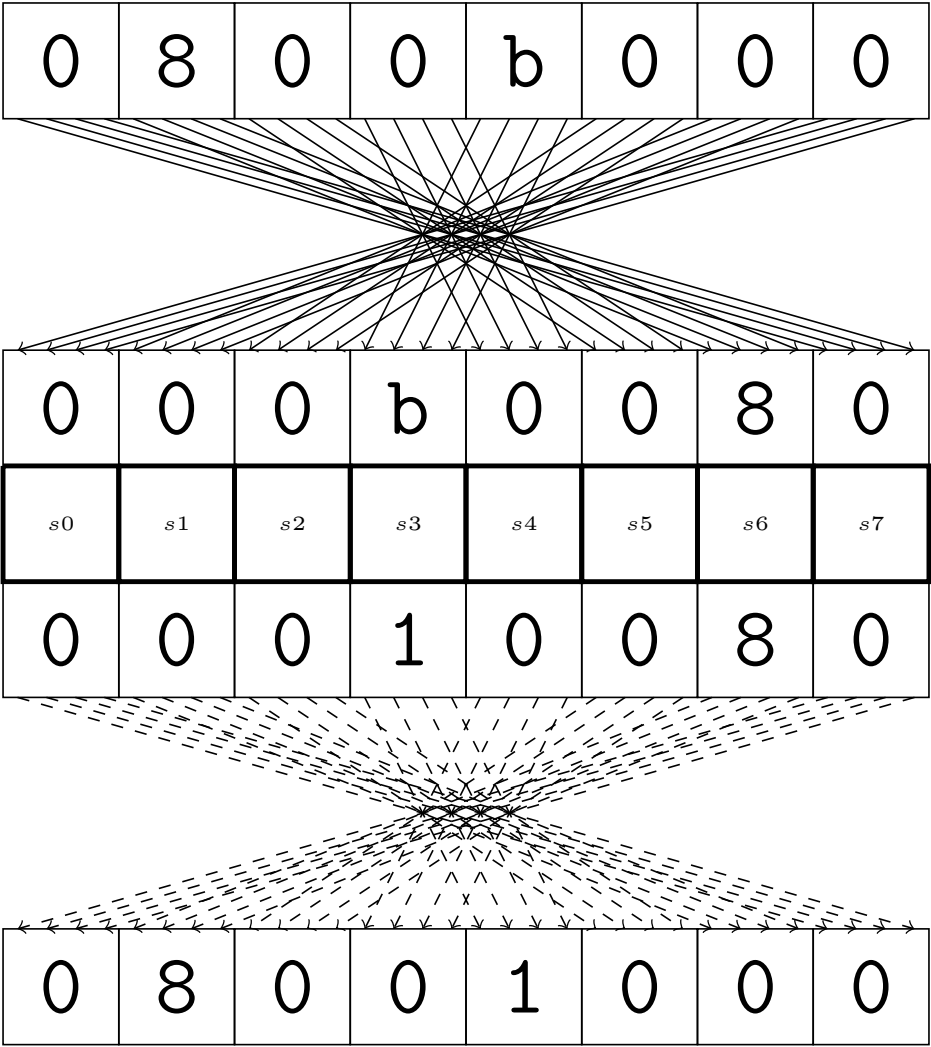
10 LBlock_round

Maximal linear correlation: $2^{-2.0}$



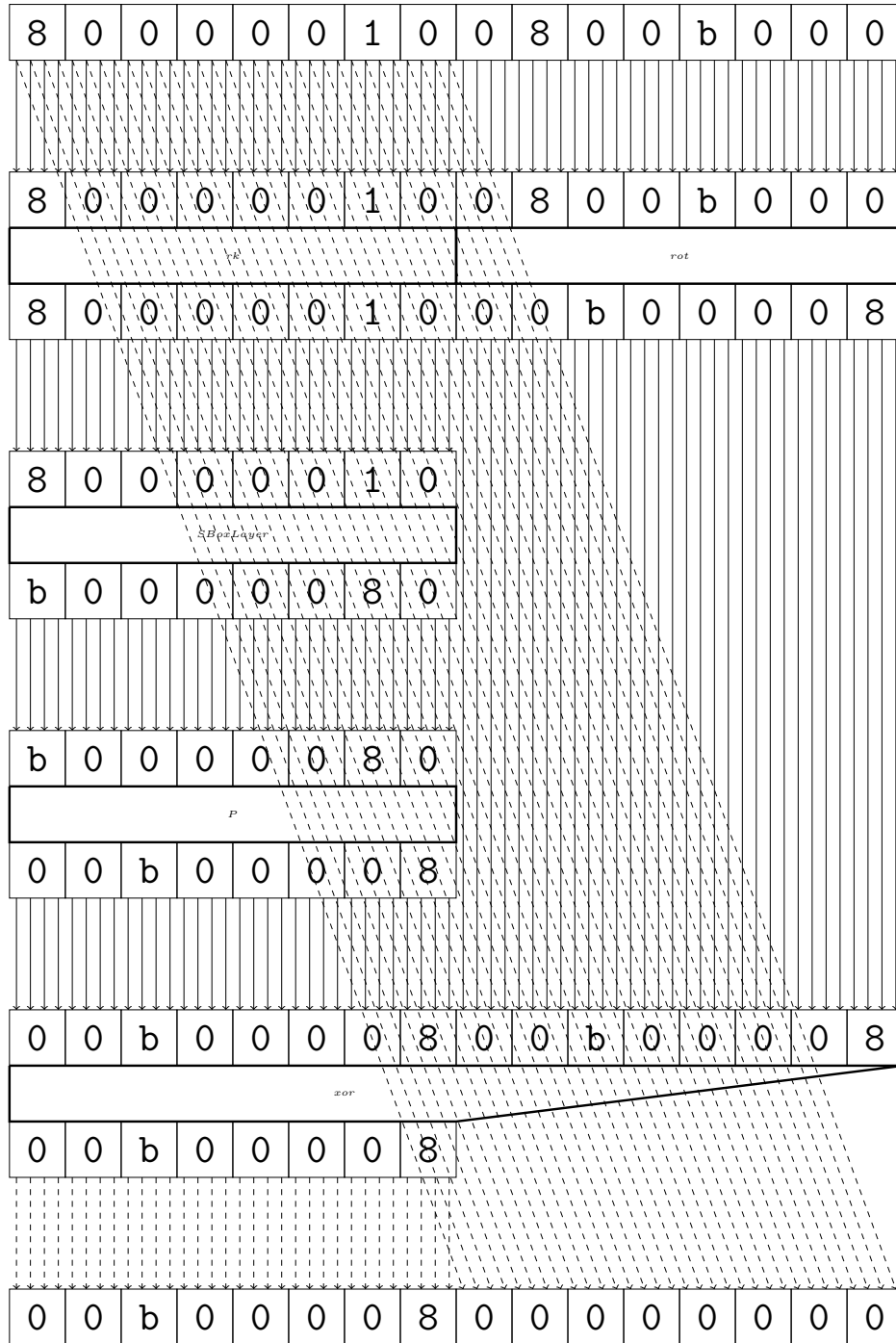
11 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



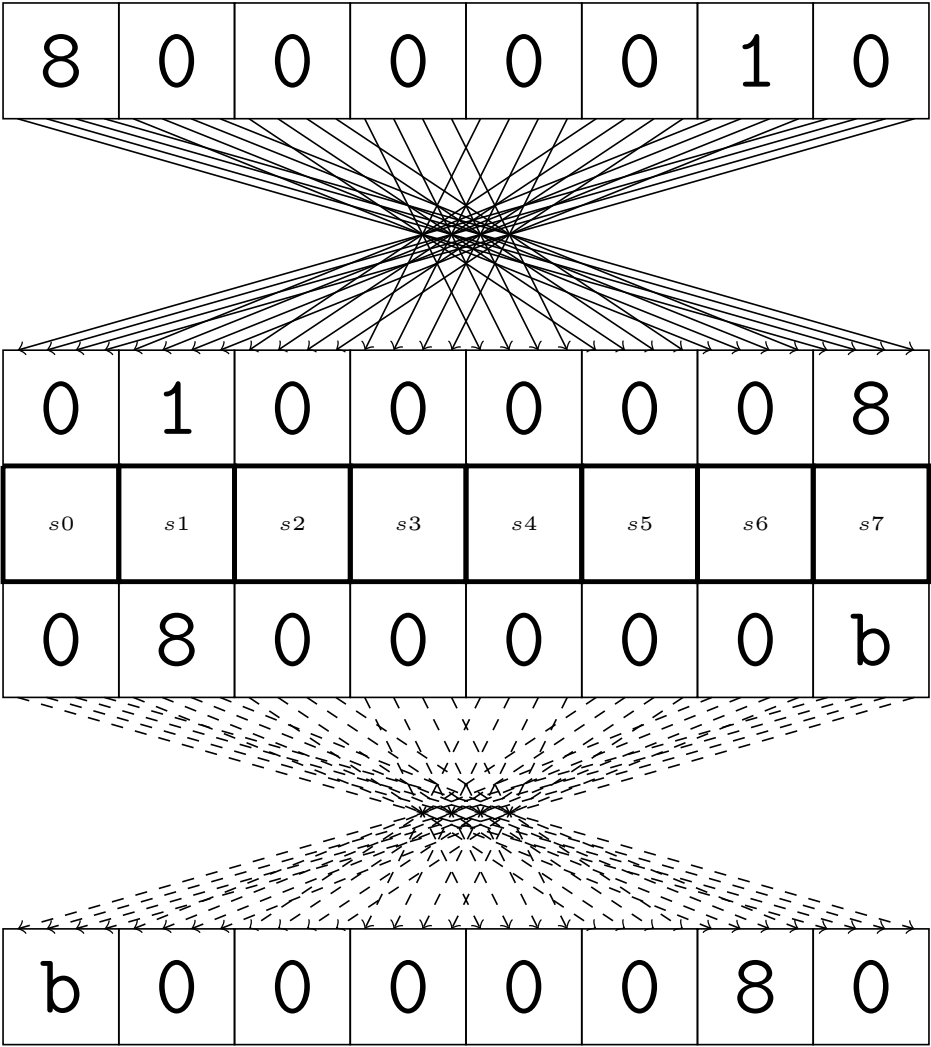
12 LBlock_round

Maximal linear correlation: $2^{-2.0}$

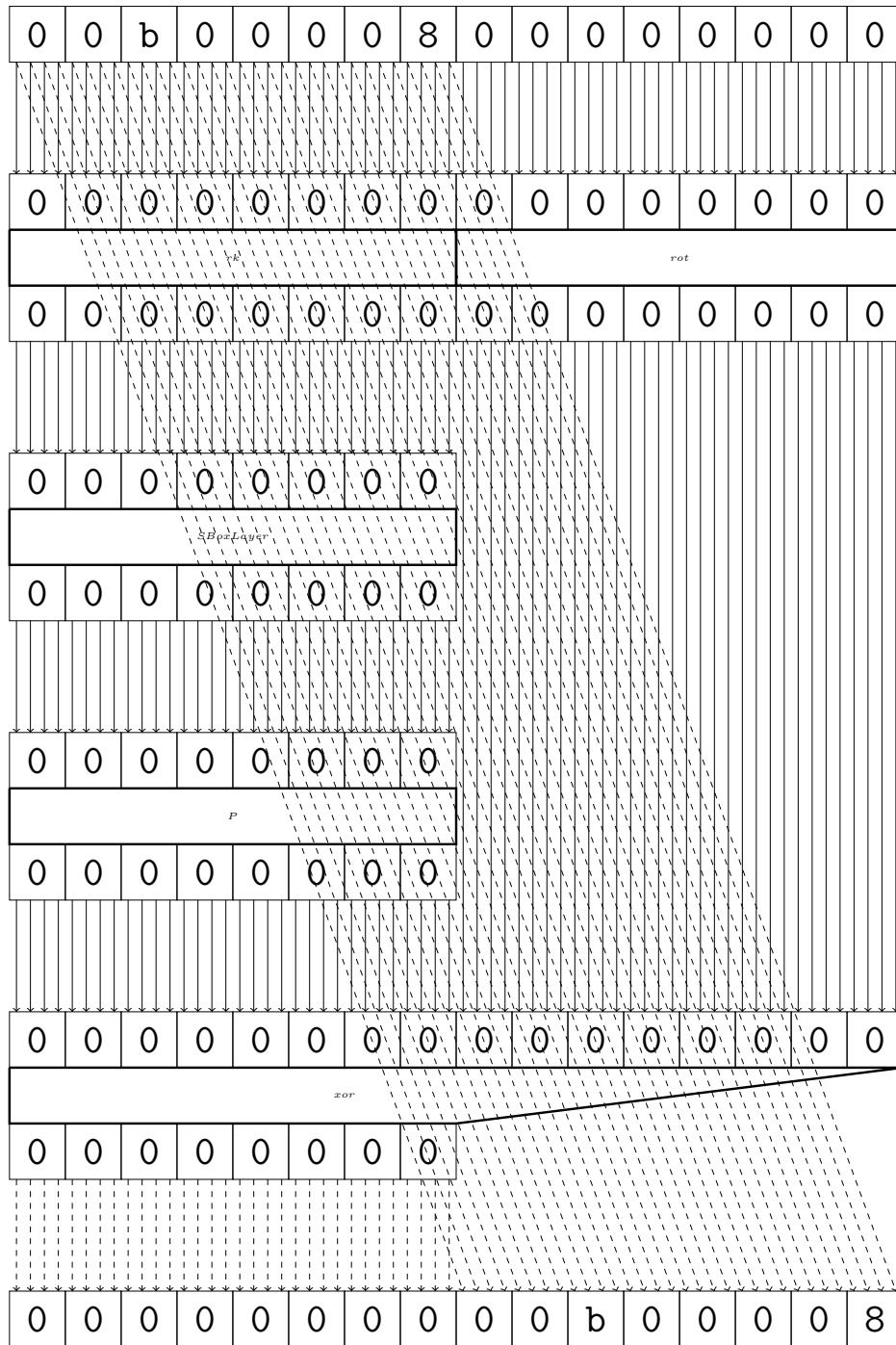


13 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

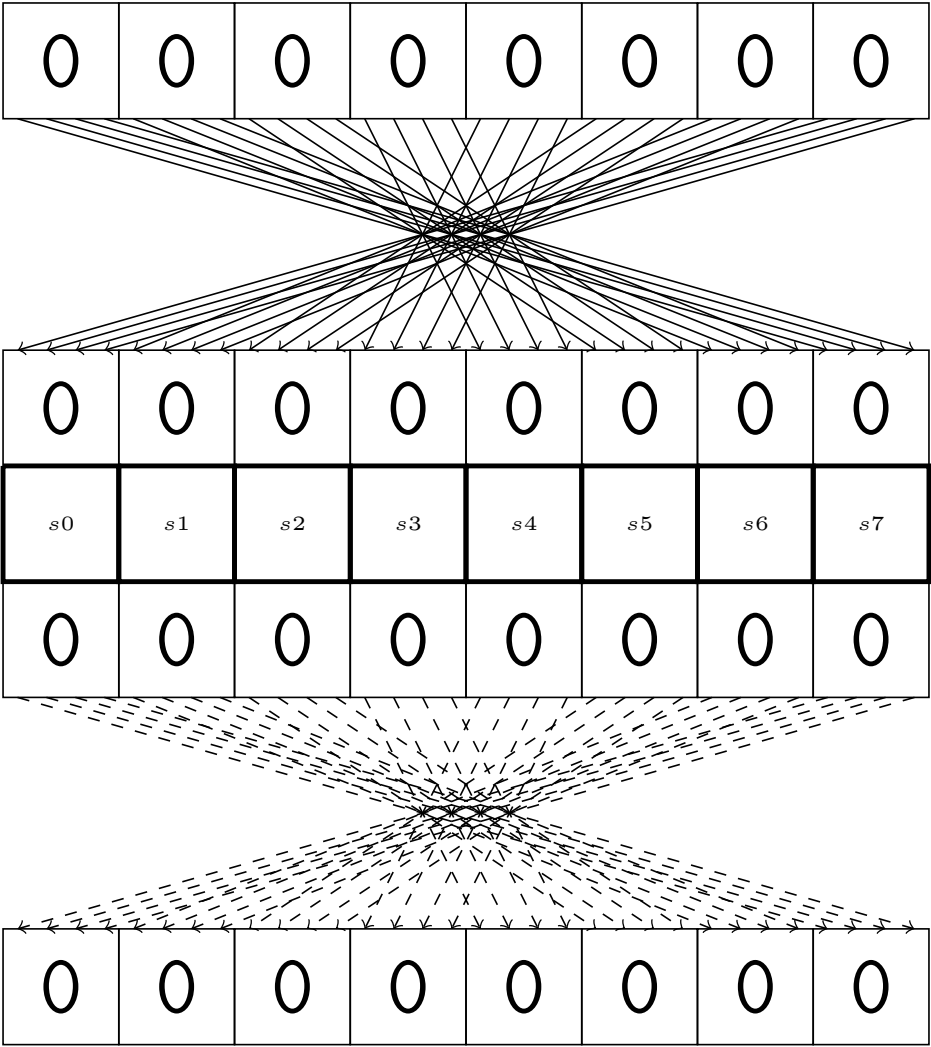


14 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

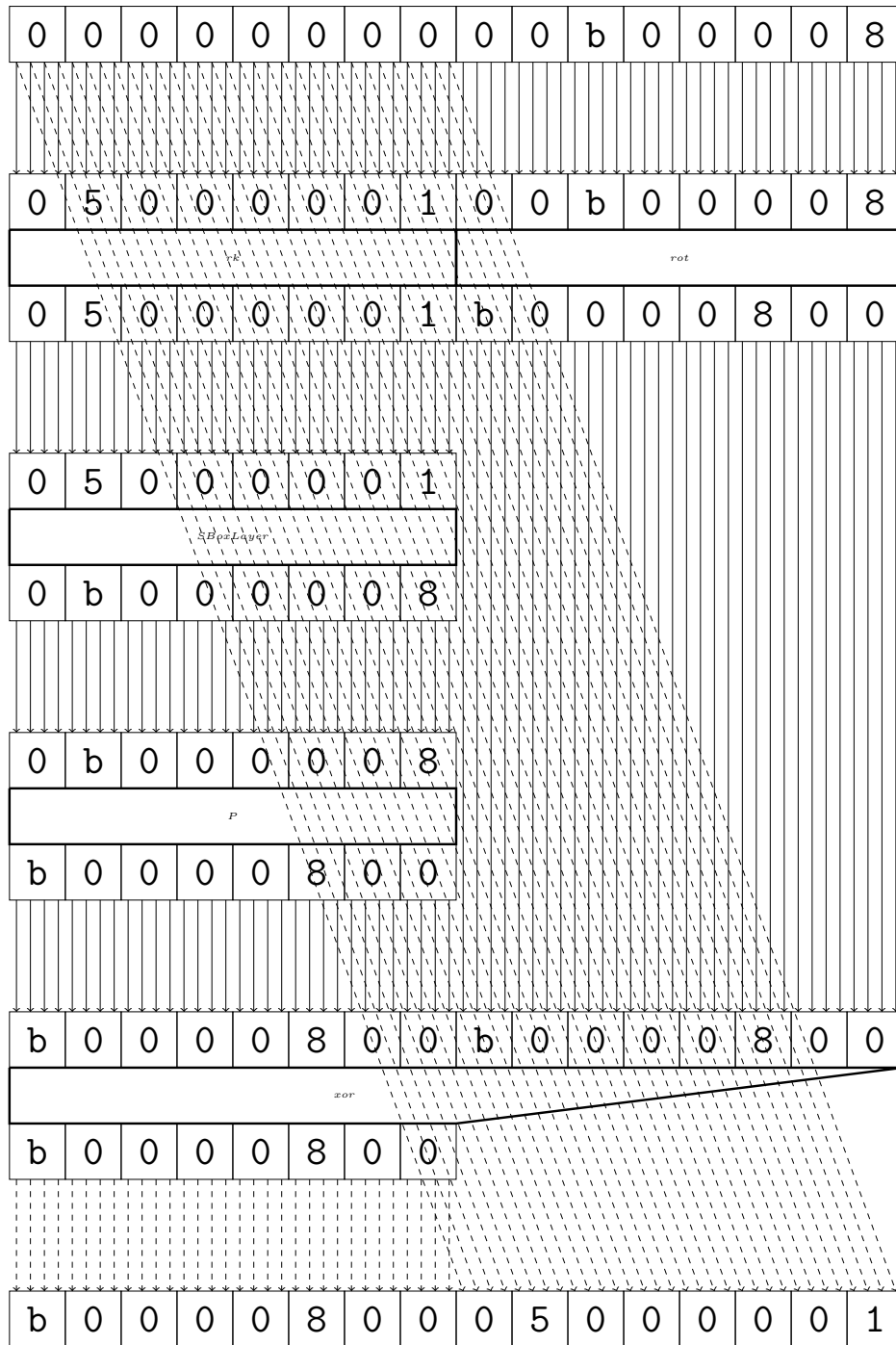
15 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



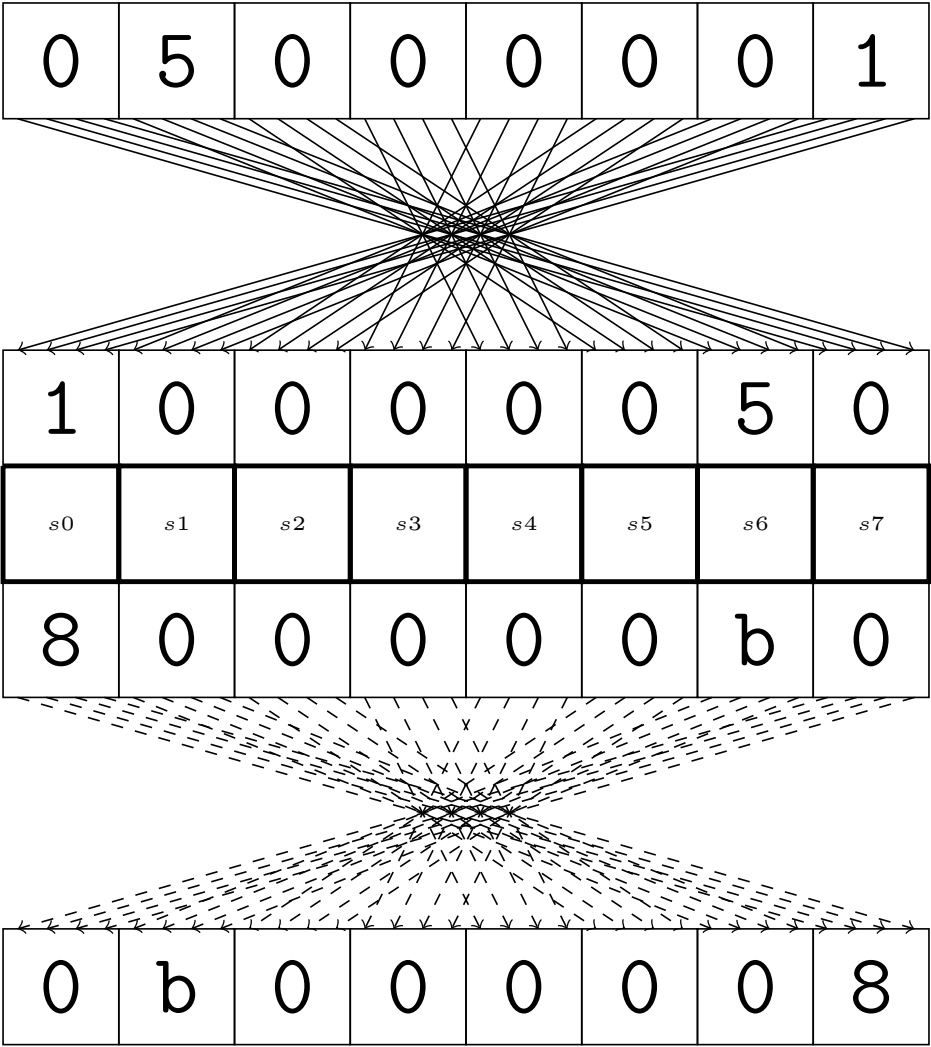
16 LBlock_round

Maximal linear correlation: $2^{-2.0}$



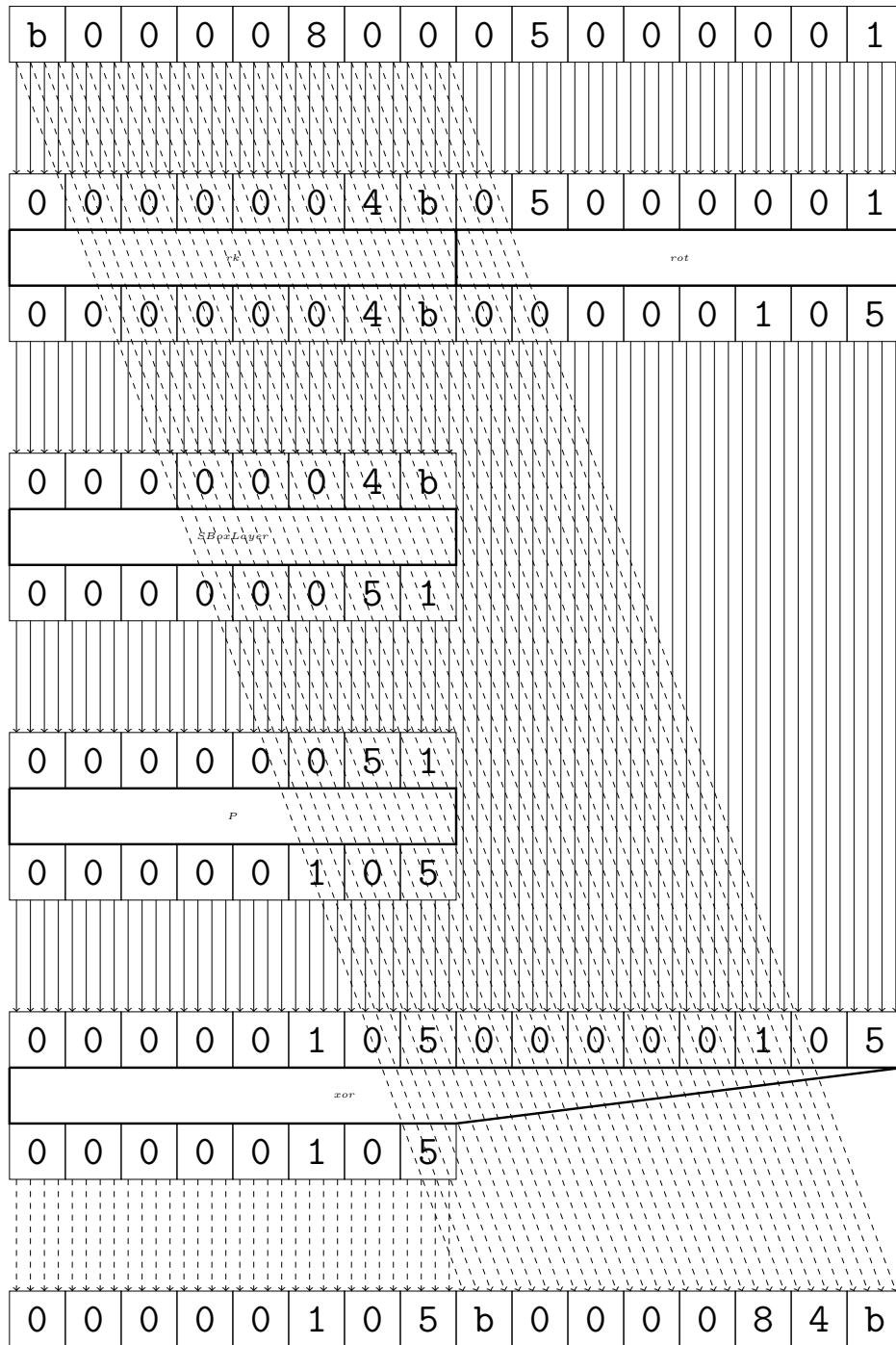
17 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



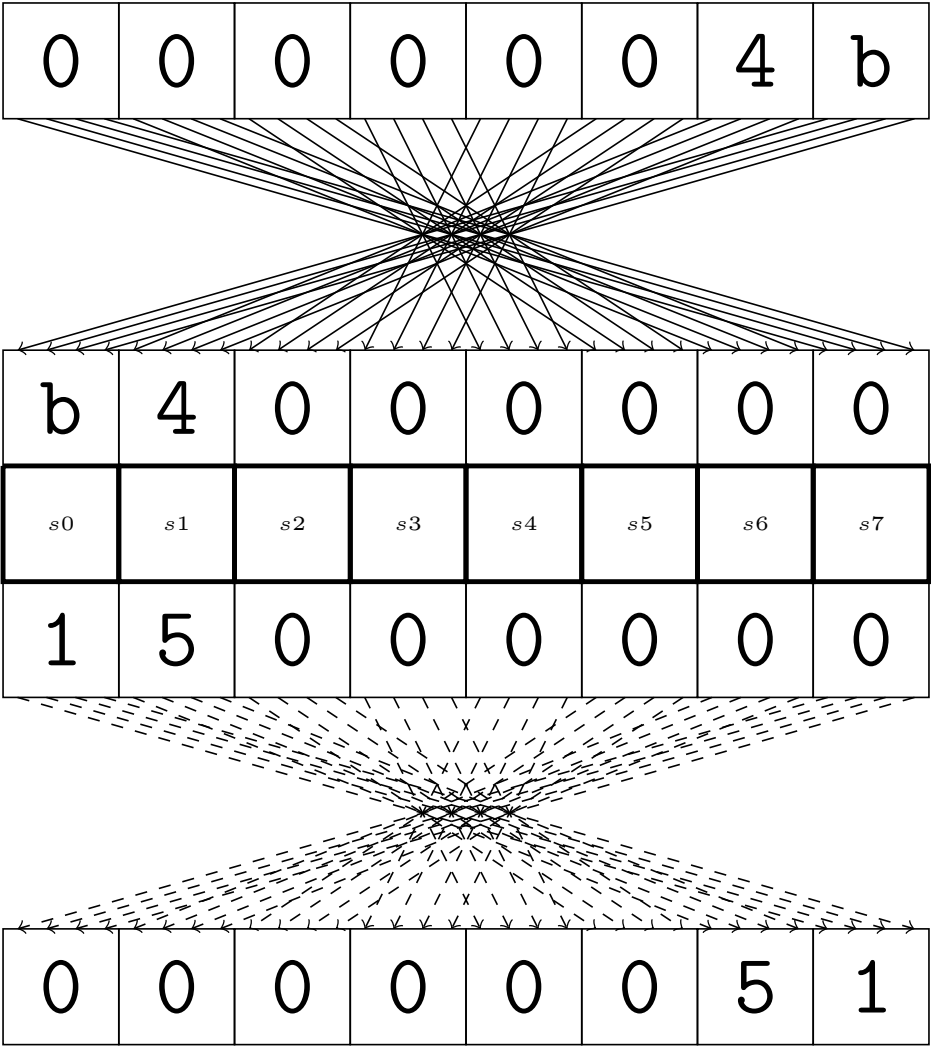
18 LBlock_round

Maximal linear correlation: $2^{-2.0}$



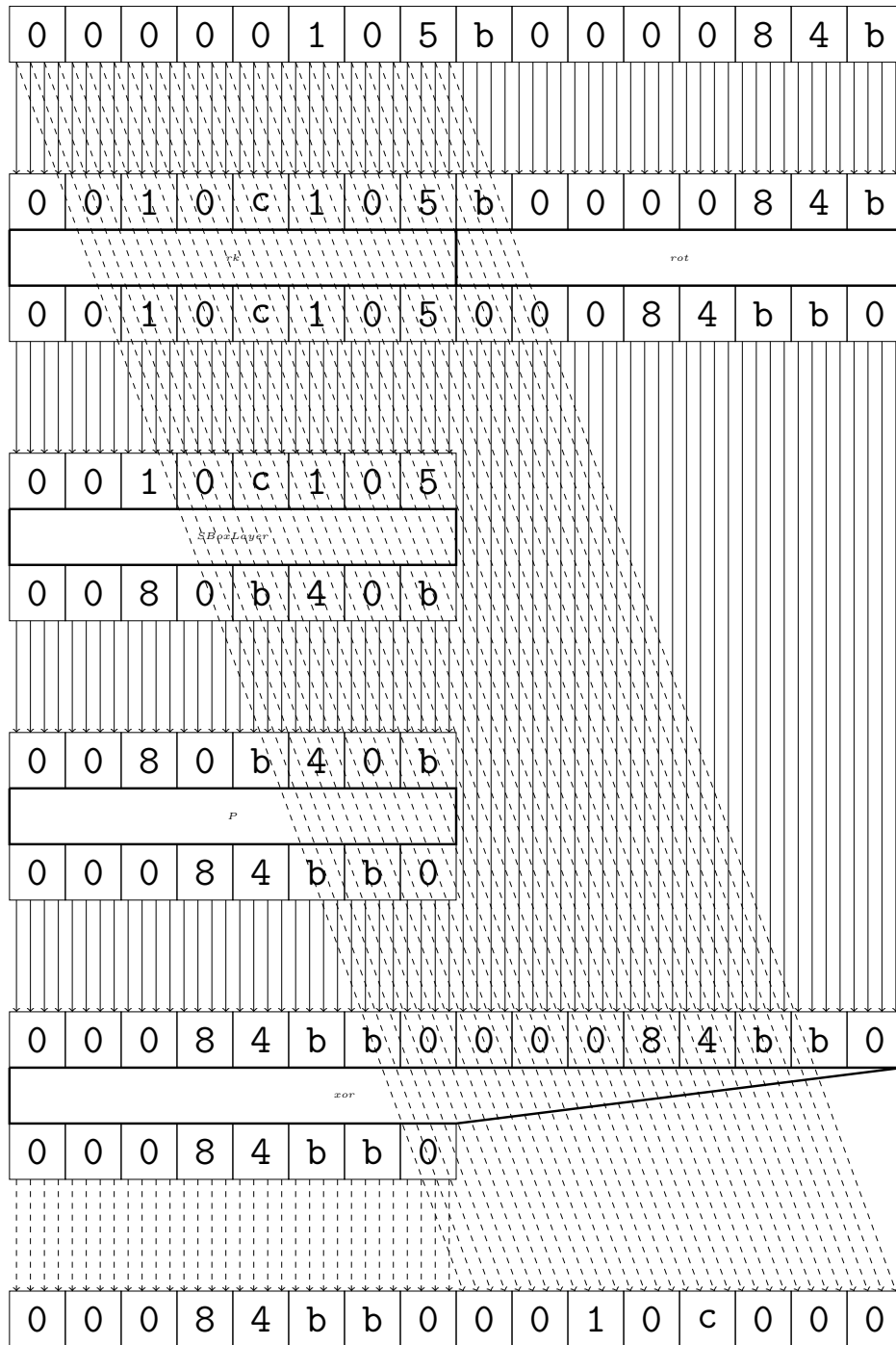
19 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



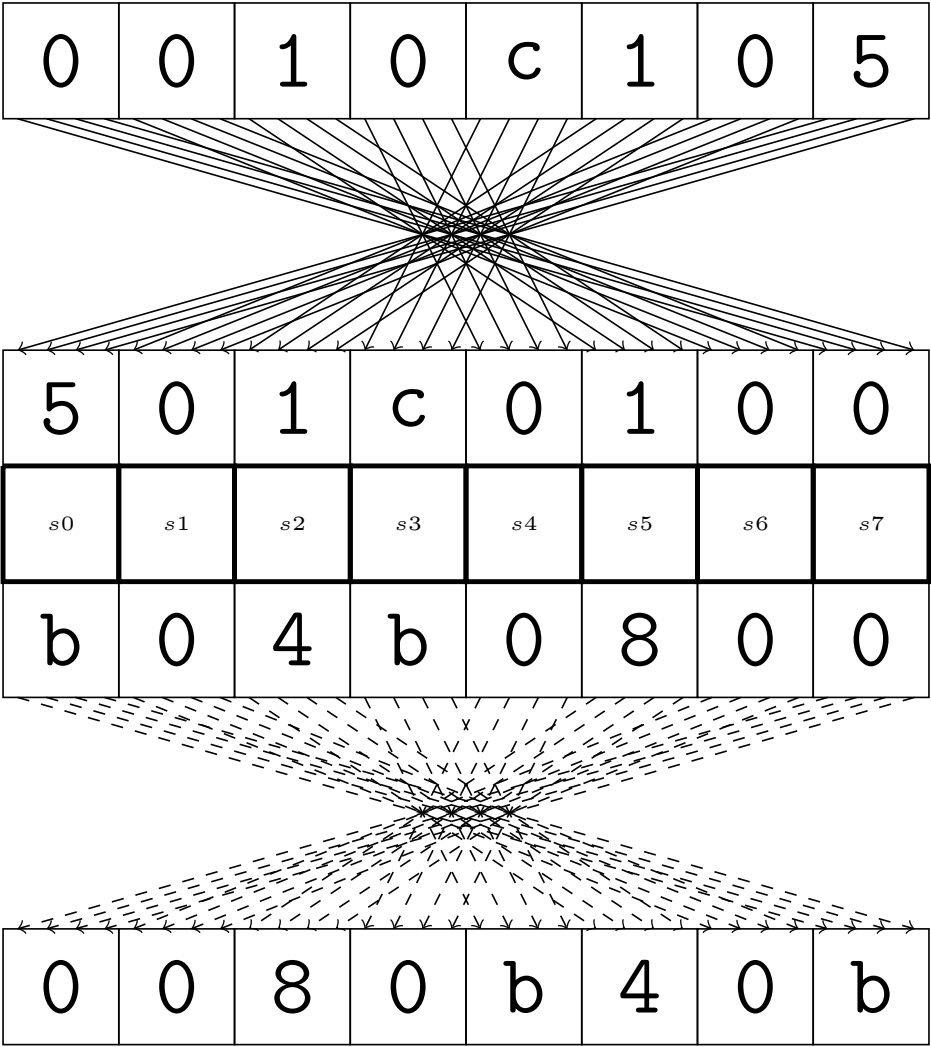
20 LBlock_round

Maximal linear correlation: $2^{-4.0}$



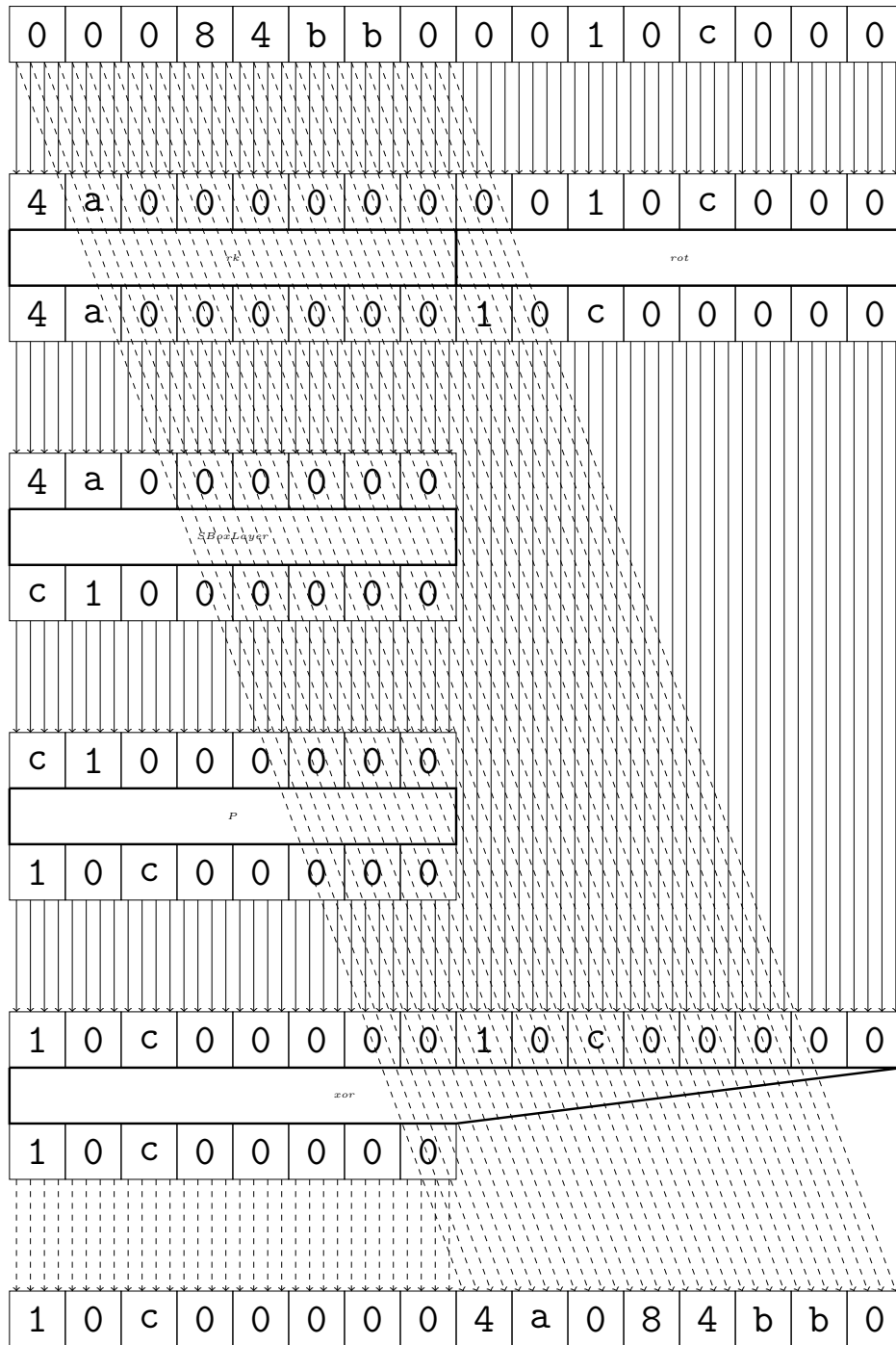
21 SBoxLayer

Maximal linear correlation: $2^{-4.0}$



22 LBlock_round

Maximal linear correlation: $2^{-2.0}$



23 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

